

Composite Shapes : MOI's and POI's

Rules for Holes



Product of Inertia, I_{xy}
Moment of Inertia $I_x + I_y$

Moment of Inertia (MOI)

Parallel Axis Thm (PAT):

$$I_x \text{ or } I_y = \bar{I} + Ad^2$$

Hole: Entire term $(\bar{I} + Ad^2)$

$$= \sum (\bar{I} + Ad^2)$$

is negative. Makes no difference

if d is + or neg from an axis because it is squared.

Product of Inertia (POI)

$$PAT = I_{xy} = \sum (\bar{I}_{xy} + Ad_x dy)$$

Hole: Entire term $(\bar{I}_{xy} + Ad_x dy)$ is negative.

Plus: (a) $\bar{I}_{xy}$ may or may not be negative

$$\bar{I}_{xy} = (-1)^n \frac{1}{72} b^2 h^2 \begin{matrix} + & - \\ \triangle & \triangle \\ - & + \end{matrix} \quad \text{or} \quad \begin{matrix} + & - \\ \square & \square \\ - & + \end{matrix} \bar{I}_{xy} = (-1)^n \cdot 0.1647 R^4$$

(b) Do not make A negative because already making the $(\bar{I}_{xy} + Ad_x dy)$ term negative.

(c) dx or dy can be negative, depending on quadrant locations.

$$\begin{array}{cc|cc} (-dx, +dy) & \square & & \square - (+dx, +dy) \\ & & x & \\ (-dx, -dy) & \square & & \square (+dx, -dy) \end{array}$$

Centroids :

$$\bar{x} = \frac{\sum \tilde{x} A_i}{\sum A_i}$$

$$\bar{y} = \frac{\sum \tilde{y} A_i}{\sum A_i}$$

Holes: A_i is negative.

$\tilde{x}_i, \tilde{y}_i$ are + or neg depending on quadrant of A_i .